is also useful in removing bacteria from heat-sensitive pharmaceutical products and biological solutions.

Membrane filters are made up of either cellulose acetate, cellulose nitrate, polycarbonate, polyvinylidene fluoride or other synthetic porous materials. These filters remove microorganisms by screening them out, such as a sieve separates large sand particles from small ones. Membranes with pore size of 0.2µm in diameter are used to remove most vegetative cells but not viruses. These filters are used to sterilize pharmaceutical products, ophthalmic solutions, culture media, oils, antibiotics, and other heat sensitive solutions (Figure 4.5a, b & c).

Depth filters

Depth filters are the oldest type of filters and they consist of overlapping layers of fibrous sheets of paper, asbestos or glass fibers. The overlapping fibers create random paths through the filter that trap many particles. Depth filter are made up of diatomaceous earth (Berkefeld filters) which are used as water purifiers. Examples of types of depth filters (Figure 4.6) contains unglazed porcelain (Chamberl and filters) and asbestos (Seitz Filter).

Air filtration

Air also can be sterilized by filtration.

HOTS

Give a reasonable method of sterilization for the following.

1. Operation theatre 2. Serum 3. Pot of soil 4. Plastic Petri Dishes 5. Rubber gloves 6. Disposable syringes. 7. Metal instruments 8. Flask of nutrient agar 9. Milk 10. Papers with spores.

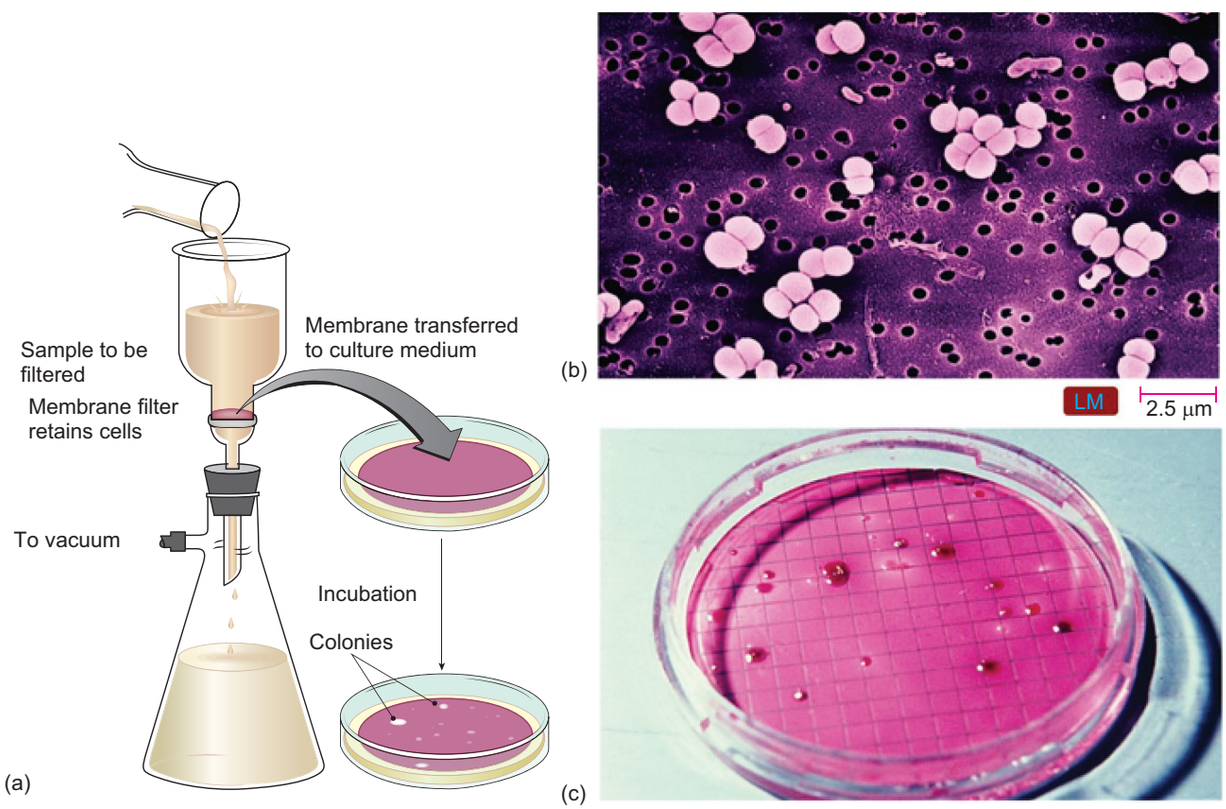


Figure 4.5: (a) Membrane filter apparatus (b) Light microscope image of microorganism filtered through membrane filter (c) Membrane filters showing microbial colonies on culture media

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